P V Narasimha Rao's son Prabhakar Rao to link Reddy and give the case a political overtone, but all these plans came unstuck.

It was not just our personal equations with Russians that tilted us towards the Glavkosmos offer; the engine on offer was unmatched in many ways. It was KVD-1, the engine that I stumbled upon at the Aerospace Exhibition in Paris in 1975. With a specific impulse of 461 seconds, this engine was far better than any American or French cryogenic engine.

There was something mysterious about KVD-1. Imagine a country making more than 250 such engines of fine caliber and stocking them! Moreover, it was something the Russians never wanted to flaunt.

In his book *Russia in Space: The Failed Frontier?* prolific space writer Brian Harvey writes about the engine:

One of the most remarkable episodes of the Russian space programme was the emergence, out of the hangar, of a secret engine developed by the (Isayev) bureau in the 1960s. In the late 1980s, India was looking for a third-stage hydrogen-powered rocket motor able to get its satellites into 24-hour orbit.

India especially wanted to orbit Gramsat, a multipurpose telecommunications satellite, to bring educational television to the villages of rural India. Knowing it would take at least 10 to 15 years to develop the technology itself, India sought help from abroad.

The engine in question was the KVD-1, built by the Isayev design bureau. No one had heard of the engine, and the West was certain that the USSR had never been able to develop cryogenic engine technology …

The KVD-1 engine has a thrust of 7,100kg, a burn time of 800 seconds and a combustion chamber pressure of 54.6 atmospheres. It weighs 282kg, is 2.1 metres tall and has a diameter of 1.6 metres.

The KVD-1 had a turbo pump-operated engine with a single fixed thrust chamber, two gimbaled thrust engines, could operate for up to 7.5 hours and be restarted five times.

… The KVD-1 had unsurpassed thrust and capabilities that made it unmatched for years. Its specific impulse of 461 seconds was still the